

**Arespectfulness:
Invisible Inheritance and the Stack of Solved
Impossibilities**

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June 2026

Abstract

Modern agents act through inheritances they did not build and seldom perceive: an oxygenated atmosphere produced by cyanobacteria, the mitochondrial machinery that supplies neural energy, controlled fire, written memory, and the compilers that turn source code into machine behavior. We name the resulting failure of perception *arespect*: the absence of felt regard for the depth, cost, and difficulty of an enabling condition, which settles once a hard-won achievement stabilizes into a reliable interface. *Arespect* differs from disrespect and from ingratitude, both of which presuppose that the thing has been perceived. Its mechanism, *stack blindness*, is the withdrawal of a working layer from attention while the agent acts through it; the layer returns to view when it fails. A comparison with disenchantment (Weber), reification (Lukács), alienation, inattentional and change blindness, the withdrawal of the ready-to-hand (Heidegger), infrastructural invisibility (Star), black-boxing (Latour), and tacit knowing (Polanyi) shows that each describes the mechanism or the content of withdrawal, and none describes the dispositional failure of regard that withdrawal permits. The stack is then traced across scales, from the Great Oxidation Event around 2.4 to 2.3 billion years ago and the endosymbiotic origin of mitochondria, through the brain's consumption of about 20% of resting energy, to fire, writing, and software abstraction, with each empirical claim graded as established, plausible inference, or live controversy. The structural claim, that inherited difficulty drops from perception because of the success of what drops, rests only on the established layers. The proposed remedy is depth perception, including an accurate estimate of how much of the depth remains unknown.

1. Introduction

Controlled fire was once a difficult competence. The hominin who first kept an ember alive through a cold night had to defend it against damp, wind, exhaustion, and the tendency of an untended fire to go out. The cooked meal, the forge, the kiln, and the glass furnace all depend on that competence, which now sits inert inside a disposable lighter. The same compression recurs at later layers. A single command to build a program depends on parsing, type theory, register allocation, an instruction set, and a long record of failed designs, and a programmer who has never written a compiler experiences none of that when typing the command. The difficulty was paid once and then hidden, and the hiding is what makes the achievement usable.

Anyone living after a long accumulation of such achievements treats the top of the stack as the ground. A reader breathes oxygen made by cyanobacteria, thinks with energy released by mitochondria, and reads an alphabet with roughly 4,000 years of history, without registering any of it. Breath, text, and a deployment to production each rest on something that was once frightening, sacred, or nearly impossible. The concealment is a consequence of leverage. A layer becomes forgettable by becoming dependable, and dependability is the property that removes a thing from notice.

The attitude this produces toward the stack lacks a name. It is seldom contempt; few people despise the power grid, the alphabet, or the chemistry of respiration. It resembles numbness, an

inability to feel the weight of what one stands on. We call it *arespect*: the absence of felt regard for inherited difficulty, where the difficulty has ceased to be perceived at all, so that no weighing or dismissal takes place. The condition is *arespectfulness*, and its mechanism, the withdrawal of a working layer from view, is *stack blindness*. We argue three things: that *arespect* is a distinct and consequential failure of perception; that it has the same form whether the inherited layer is geological, biological, or technical; and that it is produced by the success of the thing it concerns, so that respect has to be achieved against the default and is not supplied by good intentions.

2. Definition of *arespect*

Three stances toward an inherited thing can be distinguished. *Respect* perceives its value and its cost and grants it regard. *Disrespect* perceives the value and diminishes it, as when a cynic calls a cathedral a shed. *Arespect* does neither, because nothing has registered to be weighed. The decisive difference lies between the second and third stances. *Disrespect* follows perception and can be argued with: a person can be shown that the thing they belittled was worth more than they said. *Arespect* precedes perception, so argument has nothing to act on; a valuation that was never made cannot be revised. The etymology is apt. *Respect* derives from *respicere*, to look again, and *arespect* is the condition of never having looked twice, of treating a standing achievement as furniture.

Several distinctions fix the concept. *Arespect* is a disposition and not a lapse: the claim concerns a person's standing relation of unfelt reliance on the grid, and a single failure to notice the grid on a given morning would not qualify. It is evaluative without being a charge of bad character. To call a stance *arespectful* is first to say that it fails to track something real, the buried labor and solved problems that a layer contains, and only secondarily, if at all, to assign blame. A person can be entirely innocent and thoroughly *arespectful*, which places *arespect* closer to a perceptual deficit than to ingratitude. Gratitude is one possible response to perceived depth; *arespect* is the prior absence of the perception that gratitude, or its refusal, would be about. The object of the failure is also specific. The *arespectful* user of a bridge sees the bridge and crosses it; what fails to register is the bridge as an achievement, with its ore, metallurgy, load calculations, and decades of maintenance.

A new term must be distinguished from a large set of related concepts. Each names something real and adjacent, and each names either the mechanism by which a layer drops from view or the content that drops. None names the failure of regard that the dropping makes possible.

Concept	What it names	How <i>arespect</i> differs
Disenchantment (Weber [1919] 1946)	The cultural draining of mystery from a world made calculable	Removes wonder from the knowable; <i>arespect</i> removes regard for difficulty, and can coexist with full re-enchantment

Concept	What it names	How arespect differs
Reification (Lukács [1923] 1971)	Human relations appearing as relations between things	Misreads the human as thing; arespect fails to perceive the difficulty inside the thing, and persists outside any commodity form
Alienation (Marx [1844] 1959)	The producer's estrangement from their own labor and its product	Concerns the producer's own activity; arespect concerns the beneficiary's relation to others' sedimented labor
Inattentional blindness (Mack and Rock 1998; Simons and Chabris 1999)	Failure to perceive an unattended stimulus under load	A momentary, capacity-limited lapse; arespect is a settled disposition that survives full attention
Change blindness (Rensink et al. 1997)	Failure to detect a change across views	A detection failure; arespect is a valuation failure, intact perception with absent regard
Misplaced concreteness (Whitehead 1925)	Mistaking an abstraction for the concrete reality	Takes the abstract as real; arespect flattens the concrete depth of the real into a frictionless given, the inverse error
Ready-to-hand withdrawal (Heidegger [1927] 1962)	The working tool receding from notice in skilled use	Names the phenomenological mechanism; arespect is the stance that hardens once the receded is taken for empty
Infrastructural invisibility (Star 1999)	The structural property by which working infrastructure goes unseen	A property of systems; arespect relocates the phenomenon in the disposition of the perceiver
Tacit knowing (Polanyi 1966)	The structure of attending from subsidiary particulars to a focal whole	Describes a necessary and healthy reliance; arespect is the pathology of mistaking the subsidiary for the nonexistent

Every entry in the middle column describes how or why a layer leaves awareness, and in nearly every case the withdrawal is necessary, useful, and often a sign of competence; no agent could act while attending to every condition of acting. The right-hand column isolates the remainder once this is granted: the failure that occurs when a useful withdrawal is taken as a fact about the

world, when “I need not look at this to use it” becomes “there is nothing here to look at.” The new term names that transition alone. Without a term for it, the vice is repeatedly confused with the harmless mechanisms of invisibility, withdrawal, black-boxing, inattention, and reification that make it likely.

3. Mechanisms of withdrawal

Several independent traditions describe the withdrawal that makes arespect possible, and they agree on its structure. Heidegger’s analysis of equipment in *Being and Time* supplies the phenomenology. A tool in skilled use is *zuhanden*, ready-to-hand, and its readiness consists in self-effacement: the hammer is the means through which attention reaches the work, and the more fluent the use, the more completely the hammer disappears from notice (Heidegger [1927] 1962). It becomes an object again when it fails. When the head flies off or the handle splits, the hammer appears as a present-at-hand thing with weight, grain, and a price, conspicuous in proportion to its uselessness. While it works it is unseen, and breakdown returns it to view.

Susan Leigh Star extended the same structure from the hammer to the water main and gave it a sociology. Working infrastructure is embedded in other structures, learned as part of membership in a community, built on an installed base, and transparent in use, and from these properties follows the one that matters here: it becomes visible on breakdown (Star 1999; Star and Ruhleder 1996). When the server goes down, the bridge washes out, or the power fails, the substrate comes into view, often together with the maintenance arrangements that had been sustaining it unnoticed (Bowker and Star 1999). On this account invisibility is a condition of infrastructure functioning, and a road that had to be assembled deliberately for each journey would fail as a road.

Latour stated the epistemic version. Black-boxing is the way technical work is made invisible by its own success: when a machine runs efficiently or a fact is settled, attention goes only to its inputs and outputs and not to its internal complexity, so that the more science and technology succeed, the more opaque they become (Latour 1999). The opacity is produced by success and grows with it, and better explanation cannot remove it while the success continues. A technology that everyone still understood in detail would be one that had not yet become embedded enough in use to be widely useful.

Polanyi supplies the cognitive structure that connects these accounts and locates the line that arespect crosses. In any skilled act, attention proceeds *from* a set of subsidiary particulars *to* a focal whole. The pianist attends from the keys to the music and the reader from the letters to the sense, and turning focal attention back onto the particulars in the middle of a phrase disrupts the performance (Polanyi 1966). Reliance on the stack takes the form of dwelling in it and seeing through it, and this attending-from is how competence works. A demand to attend to every inherited layer at all times would be a demand for paralysis, and the concept of arespect makes no such demand. The pathology lies in a further step that the act itself does not require, from “I rightly look through this” to “there is nothing here to look at.” The step is short. A listener who hears a flawless performance and concludes that playing must be easy, with no decade of practice and no history of

instrument-building behind the music, has taken it. Respect does not ask the listener to watch the keys; it asks only that smoothness not be read as evidence of shallowness.

Two further sources describe the content that drops out. Simondon argued that culture meets technical objects with a built-in blindness to their genesis, treating them either as inert utilities or as threatening robots, and in both cases missing that a technical object is the residue of a long individuation, a gradual convergence toward internal coherence that he called concretization, with its formative history folded inside it (Simondon [1958] 2017). What disappears is that history, the ordered sequence of solved problems that made the object what it is. Stiegler extended the point to time. On his account the human is constituted by externalizing memory and experience into technical supports, which he called tertiary retention, and this inheritance reaches each newcomer already complete, with the labor of its making removed, so that forgetting is built into the channel through which technics is transmitted (Stiegler [1994] 1998). In his terms the stack is externalized memory delivered without a record of its cost. This explains why *arespect* is the default and respect the exception: mechanisms of withdrawal operate everywhere, and the content they remove is erased by the manner of its transmission.

4. Atmospheric oxygen as an inherited layer

The stack begins before tools. The oldest inherited layers are geological and biological, and they conceal their difficulty more completely than tools do, because they were laid down before anything existed that could register them. Atmospheric oxygen is the plainest case. The oxygen a person consumes at an involuntary rate is the residue of planetary-scale biological work, and to treat it as a neutral given is *arespect* toward an inheritance 2 billion years deep.

Through most of the Archean the atmosphere held almost no free oxygen, below a hundred-thousandth of the present level. Its permanent rise, the Great Oxidation Event, is placed in the early Paleoproterozoic, with onset bracketed around 2.4 to 2.3 billion years ago and a transition that continued well afterward; the rise is recorded in the disappearance of a mass-independent sulfur-isotope signature that survives only under an oxygen-poor atmosphere (Lyons et al. 2014; Bekker et al. 2004). The source was oxygenic photosynthesis in cyanobacteria. One feature of the timeline matters for the argument: the capacity to produce oxygen appears to predate its lasting accumulation by hundreds of millions of years, because reduced iron and other sinks in the oceans and crust consumed the early oxygen before it could build up in the air (Lyons et al. 2014). Recent high-resolution work places the coupled oxygenation of atmosphere and ocean near 2.3 billion years ago and emphasizes that the transition was halting and protracted (Ostrander et al. 2024). The exact onset remains disputed across a wide window, and the oxygenated atmosphere is best dated as the outcome of a slow planetary process and not as an event.

For the life already present, the change was a chemical assault. Molecular oxygen and the reactive species derived from it damage iron-sulfur clusters, iron centers, and DNA, and the chemistry of that damage is understood at the level of mechanism (Imlay 2003, 2013). Aerobic life answered with a permanent defensive apparatus of superoxide dismutases, catalases, and peroxidases, so the inheritance that now sustains life first had to be survived. The usual account overreaches at

this point. There is no direct evidence for a discrete mass extinction of anaerobes, an “oxygen catastrophe” in the sense of a documented die-off, and the microbial record of the period is too sparse to support such a claim. The evidence supports a narrower conclusion: rising oxygen imposed strong selection, drove the evolution of antioxidant machinery, and confined obligate anaerobes to the anoxic refuges they still occupy (Sessions et al. 2009). Oxygen was simultaneously a poison and a richer energy source than any before it, the basis of an aerobic metabolism many times more efficient than fermentation, and the combination of these two properties accounts for much of the planet’s transformation.

The chronology extends further back. The last universal common ancestor, the population from which all extant life descends, is reconstructed as an anaerobe, and this conclusion is secure, over-determined by the absence of free oxygen on the early Earth. The most influential reconstruction goes further, inferring from genes traceable to that node a hydrogen-dependent, carbon-fixing metabolism in a hydrothermal-vent setting (Weiss et al. 2016). That fuller portrait is a strong inference and not an established fact: it is exposed to undetected lateral gene transfer and has been revised by later work that changes the organism’s estimated age and gene count (Moody et al. 2024). The secure claim is the minimal one. The ancestor was anaerobic, and the oxygenated world its descendants now treat as the natural home of life would have been foreign and probably lethal to it.

5. Mitochondrial energy and cognition

Between atmospheric oxygen and the thought that considers it lies a third inherited layer, the one that converts the first into the second. Aerobic eukaryotic life runs on mitochondria, and their origin is among the few deep-evolutionary facts firmly established. The mitochondrion descends from a free-living alphaproteobacterium taken into an archaeal host in a single endosymbiotic event, an idea revived by Lynn Margulis in the 1960s and since placed beyond serious doubt by phylogenomics (Margulis 1967; Muñoz-Gómez et al. 2022). The order of events remains disputed: whether acquisition of the bacterium triggered the complex cell or occurred in a lineage already becoming complex. The hydrogen hypothesis, the discovery of the Asgard archaea as the closest prokaryotic relatives of eukaryotes, and the laboratory cultivation of a syntrophic Asgard archaeon all bear on the question without settling it (Martin and Müller 1998; Spang et al. 2015; Imachi et al. 2020). The inheritance is certain; its founding sequence is still being reconstructed.

The mechanism of mitochondrial energy production was established by a discovery that overturned the prevailing chemical-intermediate account and is now as secure as anything in cell biology. Peter Mitchell’s chemiosmotic hypothesis proposed that the respiratory chain pumps protons across the inner membrane to build an electrochemical gradient, the proton-motive force, which drives the synthesis of ATP, the energy currency of the cell (Mitchell 1961). The inner membrane folds into cristae, and the folding is functional: cristae shape, set by dedicated protein machinery and by rows of ATP-synthase dimers that bend the membrane along their ridges, organizes respiratory-chain supercomplexes and tunes the efficiency of energy conversion (Cogliati et al.

2013, 2016). Much of what makes a thought possible is determined in the geometry of this membrane.

A well-known argument attributes the origin of complex life itself to the mitochondrion, and it remains an open dispute. Lane and Martin proposed that prokaryotes face a bioenergetic ceiling, because they generate ATP across their outer membrane and larger cells gain diminishing returns, and that the mitochondrion removed the ceiling by internalizing energy-converting membranes, each with a small local genome, thereby supplying orders of magnitude more energy per gene and permitting the larger genomes and complexity of eukaryotes (Lane and Martin 2010). The mechanism is plausible, but its central quantity has been challenged directly: Lynch and Marinov recomputed the energetic cost of a gene against cell volume in both domains and found no decisive mitochondrial advantage, since eukaryotes also gave up surface bioenergetics (Lynch and Marinov 2015), and others have questioned the energy-first account of eukaryogenesis altogether (Booth and Doolittle 2015). The proposition that mitochondria made complexity possible therefore remains an open hypothesis and is used here only as one.

The version of the inheritance the evidence supports connects cognition to mitochondria through energy budgets. The brain is about 2% of body mass and consumes on the order of 20% of the body's resting energy, a share that changes little between demanding tasks and rest (Raichle and Gusnard 2002). Most of this expenditure goes to signaling, the generation of action potentials and the restoration of the ion gradients they dissipate, and detailed budgets assign the largest shares of grey-matter energy use to action potentials and synaptic currents (Attwell and Laughlin 2001; Harris et al. 2012). ATP-driven pumps do this work, and at the synapse the ATP comes from mitochondria, which are transported to presynaptic terminals to supply it and to buffer calcium, and whose local reserves support the protein synthesis required for lasting synaptic change (Devine and Kittler 2018; Rangaraju et al. 2019). The same demand makes neurons vulnerable when the machinery fails, which is why mitochondrial dysfunction recurs in neurodegeneration and why chronic stress alters mitochondrial state, although the direction of causation in disease is unresolved (Picard and McEwen 2018).

The defensible claim is limited. Cognition rests on a bioenergetic infrastructure, and mitochondrial ATP supply and calcium handling are conditions without which neural computation degrades. The evidence does not support the further claim that cristae structure or proton flux produces or explains conscious experience. Researchers who take brain energetics most seriously describe energy as a precondition for consciousness and not a correlate of it, and leave the explanatory work to theories of neural integration (Chen and Zhang 2021). One terminological confusion needs flagging. The "free energy" of the free-energy principle in theoretical neuroscience is an information-theoretic quantity, a bound on surprise under a generative model, and has no direct relation to the thermodynamic free energy a mitochondrion supplies (Friston 2010); moving between the two senses is a category error. The open question lies in interoception, the brain's representation of the body's physiological state, which is closely tied to metabolic regulation and is a candidate basis for the felt sense of being a living body (Craig 2002; Barrett and Simmons 2015), and which remains an open research frontier. A thinking subject depends on an energy infrastructure it did not build

and cannot switch off; the claim that the crista does the thinking is unsupported, and the argument here relies only on the first of these statements.

6. Fire, writing, and cumulative culture

Above the biological layers the specifically human stack begins, and it repeats the pattern in a record that can in principle be excavated and dated. Fire is the transition point. Cooking moved part of digestion outside the body: heat gelatinizes starch, denatures protein, and kills pathogens, so that more energy is extracted from food at lower cost, and on the cooking hypothesis this surplus allowed the human brain, the expensive organ of the previous section, to enlarge while the gut shrank (Wrangham 2009; Wrangham et al. 1999; Aiello and Wheeler 1995). The hypothesis is contested on timing, the point on which it most depends. Secure traces of controlled fire are old but sparse, with burned bone and ashed plant remains at Wonderwerk Cave near 1 million years ago and a hearth-bearing site at Gesher Benot Ya'akov around 790,000 years ago (Berna et al. 2012; Goren-Inbar et al. 2004), yet evidence for *habitual* fire, reliable enough to sustain an evolutionary dependency, becomes continuous only around 400,000 to 300,000 years ago, later than the hypothesis requires (Roebroeks and Villa 2011). The strict brain-versus-gut trade-off has also been weakened by work finding no simple inverse relation between brain size and digestive-tissue mass across mammals, primates included (Navarrete et al. 2011). The structural claim survives these challenges: a metabolic function was externalized into a technique, the technique became a precondition for the species that uses it, and the precondition is now hidden in the unremarkable act of eating cooked food.

Writing externalized memory, and it began as bookkeeping. In the Mesopotamian sequence the earliest writing developed from small clay tokens used to count goods, which gave way to marks impressed in clay and then to cuneiform, so that the medium later used for scripture and law originated in the recording of debts and grain (Schmandt-Besserat 1992). On the strong reading from studies of orality and literacy, writing provided memory located outside the body, durable, transmissible across distance and death, and open to the kinds of inspection that lists, contracts, and tables permit (Ong 1982; Goody 1977; Goody and Watt 1963). The stronger claim, that literacy restructured thought itself and separated literate from oral minds along a clean line, is disputed: the so-called Great Divide has been challenged by the argument that the effects of literacy belong to the particular social practices through which it is acquired and are not uniform consequences of the technology (Street 1984).

The earliest recorded critique of writing describes *arespect* at the moment of installation. In Plato's *Phaedrus* the god Theuth offers writing to the king as a remedy for forgetting, and the king replies that it will produce forgetting instead, since those who learn it will rely on external marks and cease to remember from within (Plato 1995). The complaint records a vivid perception of a layer at the moment it is installed, before it recedes. 24 centuries later the alphabet is so completely inherited that a reader passes from letters to meaning without any awareness of the abstraction phonetic writing required. The forgetting the king predicted concerned the content of memory; the forgetting that actually occurred concerns the medium itself.

That no individual could rebuild this stack is an empirical finding. Human capability is the product of a cultural ratchet, a transmission faithful enough to retain improvements and accumulate them across generations beyond what any single mind could invent (Tomasello 1999), and the resulting dependence is complete, as its failures show. Well-supplied explorers have starved in environments that sustained the people native to them, because survival there depended on a large body of local knowledge, such as which plants must be leached of toxins and how, that the explorers lacked and could not derive in the time available (Henrich 2016). That knowledge was itself a stack, inherited by local populations and as invisible to them as any stack is to its users, and its absence was fatal. The failure was not one of intelligence. The explorers were capable and still died, are-spectful in the sense that the depth of an inherited competence became perceptible to them only through its absence.

7. Abstraction in computing

In computing the concealment of difficulty under an interface is an explicit design goal and not only a side effect of success. The history hidden by the command to compile a program is recent enough to be documented. When Grace Hopper built the first compiling systems in the early 1950s, the idea that a machine could translate human-readable instructions into its own code met disbelief, on the assumption that programming was clerical arithmetic the computer could not be trusted to perform for itself (Hopper 1952). When the FORTRAN team shipped a compiler a few years later whose output was competitive with hand-written assembly, it did so against a settled conviction that automatic translation would always be too inefficient for practical use (Backus et al. 1957). The conviction was reasonable and wrong, and its defeat laid down a layer. Programmers since have inherited the compiler as a given, with no experience of the doubt that had to be overcome to build it.

The technical stack differs from the others in that its opacity is engineered deliberately, and the engineering has a literature. Dijkstra answered the charge that abstraction is a form of vagueness by reversing it: the purpose of abstraction is to create a new semantic level at which one can be absolutely precise, a level that supports reasoning without descent into the one below (Dijkstra 1972). Parnas made the idea a design discipline under the name information hiding, the deliberate sealing of a module's internals behind an interface so that the rest of the system can use the module without knowing, or depending on, how it works (Parnas 1972). Information hiding is stack blindness written into a specification, with the inability to see inside as a design requirement. Simon had supplied the general rationale a decade earlier: complex systems tend to be hierarchic and nearly decomposable, built from layered subsystems with dense internal interactions and sparse interactions between layers, and in his parable of two watchmakers, the one who builds from stable sub-assemblies finishes while the one who must hold the whole mechanism together at once does not (Simon 1962). Building complex systems requires layers, and layers create regions that nobody inspects.

The technical stack thus parallels the biological one, with one difference: the compiler was designed to conceal its operation, whereas the mitochondrion was designed by no one. For the user

the result is the same, a dependable interface that is looked through, and the engineered case exhibits the breakdown-reveals-the-stack pattern under controlled conditions. Practitioners record it as the law of leaky abstractions: every non-trivial abstraction leaks, and the hidden substrate reappears at the edges, in a performance cliff or an unintelligible error that forces the programmer down a level (Spolsky 2002). The leak corresponds to Heidegger’s broken hammer and Star’s failed server, the moment at which the black box opens and shows its contents. The essential difficulty of a problem persists beneath the conveniences built over it, as in the field’s long-standing distinction between complexity inherent in a problem and complexity added by poor tools (Brooks 1987). Artificial intelligence is the newest layer of this kind, a statistical compression of a large body of inherited human culture into a system that can be queried in a sentence. It invites an especially complete *arespect*, since the user attends to a fluent reply while the training corpus, the human labor it contains, and the machinery of compression all drop from view, and the resulting black box is opaque even to its builders.

8. Evidential status of the empirical claims

An argument about the failure to perceive depth has to state the depth of its own claims accurately, including where they are shallow. The empirical support is uneven, and the table below grades each claim, lists its sources, and names the stronger version that the argument does not rely on.

Claim used in the argument	Status	Key sources	Overclaim not relied on
Breathable oxygen is the accumulated product of cyanobacterial photosynthesis, risen permanently around 2.4 to 2.3 Ga	Established as to cause; the precise onset is contested across a wide window	Lyons et al. (2014); Bekker et al. (2004); Ostrander et al. (2024)	That the rise was a single datable event and not a protracted transition
Rising oxygen chemically threatened anaerobic life and selected for antioxidant defenses	Mechanism established; the scale of any die-off is not	Imlay (2003, 2013); Sessions et al. (2009)	That the event was a documented mass extinction, an “oxygen catastrophe”
LUCA was anaerobic	Secure; over-determined by an anoxic early Earth	Weiss et al. (2016); Moody et al. (2024)	That its hydrothermal, hydrogen-dependent metabolism is settled fact
Mitochondria arose once by endosymbiosis of an alphaproteobacterium	Established	Margulis (1967); Muñoz-Gómez et al. (2022)	That the early-versus-late timing of the event is resolved

Claim used in the argument	Status	Key sources	Overclaim not relied on
ATP synthesis proceeds by chemiosmosis across folded cristae	Established	Mitchell (1961); Cogliati et al. (2013, 2016)	That cristae geometry has any settled relation to conscious experience
Mitochondria removed a bioenergetic ceiling and so enabled eukaryotic complexity	Live controversy	Lane and Martin (2010) versus Lynch and Marinov (2015); Booth and Doolittle (2015)	That the energy-per-gene thesis is established; it must be paired with its rebuttal
Cognition depends on mitochondrial ATP and calcium handling as enabling conditions	Established as enabling-condition	Attwell and Laughlin (2001); Rangaraju et al. (2019); Devine and Kittler (2018)	That bioenergetics constitutes or produces consciousness
Cooking externalized digestion and helped permit brain expansion	Plausible inference; the timing is contested	Wrangham (2009); Aiello and Wheeler (1995) versus Roebroeks and Villa (2011); Navarrete et al. (2011)	That habitual fire is securely old enough to have driven the expansion
Writing externalized memory and began as accounting	Established	Schmandt-Besserat (1992); Ong (1982)	That literacy uniformly restructures consciousness along a Great Divide

The structural claim, that inherited difficulty drops from perception and that the dropping is produced by the success of what drops, does not depend on any single contested fact, because the established layers exhibit it as clearly as the disputed ones. A reader who rejects the thesis that mitochondria caused complex life loses nothing from the argument, since the chemiosmotic machinery is an unfelt inheritance regardless of the dispute over its consequences. The same point answers the concern that the biology carries rhetorical weight it cannot bear. The argument rests on the enabling-condition claim, which is well supported, and explicitly declines the constitutive claim that would locate the mind in a membrane; the contested propositions are marked as contested and carry no load. Perceiving the stack accurately includes perceiving how much of it remains unknown, and a respect that waited for the depths to be fully charted would never be exercised.

9. Conclusion

The same pattern appears at every scale examined. A difficult achievement, whether the oxygenation of a planet, the capture of a bacterium that became an organelle, the maintenance of fire, the externalization of memory in clay, or the automatic translation of programs into machine behavior, becomes a dependable layer. The layer is looked through; the looking-through, which is proper and necessary, hardens into the belief that nothing lies beneath; and the achievement passes to the next inheritor as a frictionless given. On this reading civilization is an accumulation of formerly impossible achievements made usable, each buried under the interface its solution produced, and *arespect* is the standing relation of those near the top of the stack to the depth beneath them.

The relation has practical costs. The blindness that hides a layer's genesis also hides the people who keep the layer working. Those who repair and maintain infrastructure work under a structural invisibility that the emphasis on innovation deepens, since successful maintenance is silent and only failure is noticed (Russell and Vinsel 2016; Vinsel and Russell 2020). A history of technology organized around use instead of novelty shows the scale of the omission: maintenance accounts for far more of the world's technical effort than invention and receives a small fraction of the attention (Edgerton 2007). *Arespect* toward the stack is therefore continuous with *arespect* toward the people who sustain it, and perceiving a bridge as a sustained achievement includes perceiving the crews who sustain it.

The remedy is neither gratitude, which can be performed without perceiving anything, nor a reverence that would require attending to the keys in the middle of a phrase. No unmediated ground exists to return to; if the human is constituted by its technical inheritance, the available choice is between meeting that inheritance with regard and meeting it without (Stiegler [1994] 1998). What remains is depth perception: continuing to act fluently through the stack while keeping its depth in view, in the literal sense of *respicere*. Because every layer of the stack is built to make itself unnecessary to see, this perception has to be cultivated deliberately. It consists in knowing that breathable air derives from cyanobacteria, that thought is paid for in proton gradients, that written words are an externalized memory whose introduction was distrusted, and that ordinary actions rest on difficulties that organisms, lineages, and a planet once had to survive.

References

- Aiello, Leslie C., and Peter Wheeler. 1995. "The Expensive-Tissue Hypothesis: The Brain and the Digestive System in Human and Primate Evolution." *Current Anthropology* 36 (2): 199–221. <https://doi.org/10.1086/204350>.
- Attwell, David, and Simon B. Laughlin. 2001. "An Energy Budget for Signaling in the Grey Matter of the Brain." *Journal of Cerebral Blood Flow & Metabolism* 21 (10): 1133–45. <https://doi.org/10.1097/00004647-200110000-00001>.
- Backus, J. W., H. Stern, I. Ziller, et al. 1957. "The FORTRAN Automatic Coding System." *Proceedings of the Western Joint Computer Conference: Techniques for Reliability*, 188–98. <https://doi.org/10.1145/1455567.1455599>.

- Barrett, Lisa Feldman, and W. Kyle Simmons. 2015. "Interceptive Predictions in the Brain." *Nature Reviews Neuroscience* 16 (7): 419–29. <https://doi.org/10.1038/nrn3950>.
- Bekker, A., H. D. Holland, P.-L. Wang, et al. 2004. "Dating the Rise of Atmospheric Oxygen." *Nature* 427 (6970): 117–20. <https://doi.org/10.1038/nature02260>.
- Berna, Francesco, Paul Goldberg, Liora Kolska Horwitz, et al. 2012. "Microstratigraphic Evidence of in Situ Fire in the Acheulean Strata of Wonderwerk Cave, Northern Cape Province, South Africa." *Proceedings of the National Academy of Sciences* 109 (20): E1215–20. <https://doi.org/10.1073/pnas.1117620109>.
- Booth, Austin, and W. Ford Doolittle. 2015. "Eukaryogenesis, How Special Really?" *Proceedings of the National Academy of Sciences* 112 (33): 10278–85. <https://doi.org/10.1073/pnas.1421376112>.
- Bowker, Geoffrey C., and Susan Leigh Star. 1999. *Sorting Things Out: Classification and Its Consequences*. MIT Press.
- Brooks, Frederick P. 1987. "No Silver Bullet: Essence and Accidents of Software Engineering." *Computer* 20 (4): 10–19. <https://doi.org/10.1109/mc.1987.1663532>.
- Chen, Yali, and Jun Zhang. 2021. "How Energy Supports Our Brain to Yield Consciousness: Insights From Neuroimaging Based on the Neuroenergetics Hypothesis." *Frontiers in Systems Neuroscience* 15: 648860. <https://doi.org/10.3389/fnsys.2021.648860>.
- Cogliati, Sara, Jose A. Enriquez, and Luca Scorrano. 2016. "Mitochondrial Cristae: Where Beauty Meets Functionality." *Trends in Biochemical Sciences* 41 (3): 261–73. <https://doi.org/10.1016/j.tibs.2016.01.001>.
- Cogliati, Sara, Christian Frezza, Maria Eugenia Soriano, et al. 2013. "Mitochondrial Cristae Shape Determines Respiratory Chain Supercomplexes Assembly and Respiratory Efficiency." *Cell* 155 (1): 160–71. <https://doi.org/10.1016/j.cell.2013.08.032>.
- Craig, A. D. 2002. "How Do You Feel? Interoception: The Sense of the Physiological Condition of the Body." *Nature Reviews Neuroscience* 3 (8): 655–66. <https://doi.org/10.1038/nrn894>.
- Devine, Michael J., and Josef T. Kittler. 2018. "Mitochondria at the Neuronal Presynapse in Health and Disease." *Nature Reviews Neuroscience* 19 (2): 63–80. <https://doi.org/10.1038/nrn.2017.170>.
- Dijkstra, Edsger W. 1972. "The Humble Programmer." *Communications of the ACM* 15 (10): 859–66. <https://doi.org/10.1145/355604.361591>.
- Edgerton, David. 2007. *The Shock of the Old: Technology and Global History Since 1900*. Oxford University Press.
- Friston, Karl. 2010. "The Free-Energy Principle: A Unified Brain Theory?" *Nature Reviews Neuroscience* 11 (2): 127–38. <https://doi.org/10.1038/nrn2787>.

- Goody, Jack. 1977. *The Domestication of the Savage Mind*. Cambridge University Press.
- Goody, Jack, and Ian Watt. 1963. "The Consequences of Literacy." *Comparative Studies in Society and History* 5 (3): 304–45. <https://doi.org/10.1017/s0010417500001730>.
- Goren-Inbar, Naama, Nira Alperson, Mordechai E. Kislev, et al. 2004. "Evidence of Hominin Control of Fire at Gesher Benot Ya'aqov, Israel." *Science* 304 (5671): 725–27. <https://doi.org/10.1126/science.1095443>.
- Harris, Julia J., Renaud Jolivet, and David Attwell. 2012. "Synaptic Energy Use and Supply." *Neuron* 75 (5): 762–77. <https://doi.org/10.1016/j.neuron.2012.08.019>.
- Heidegger, Martin. (1927) 1962. *Being and Time*. Translated by John Macquarrie and Edward Robinson. Harper & Row.
- Henrich, Joseph. 2016. *The Secret of Our Success: How Culture Is Driving Human Evolution, Domesticating Our Species, and Making Us Smarter*. Princeton University Press. <https://doi.org/10.1515/9781400873296>.
- Hopper, Grace Murray. 1952. "The Education of a Computer." *Proceedings of the 1952 ACM National Meeting (Pittsburgh)*, 243–49. <https://doi.org/10.1145/609784.609818>.
- Imachi, Hiroyuki, Masaru K. Nobu, Nozomi Nakahara, et al. 2020. "Isolation of an Archaeon at the Prokaryote–Eukaryote Interface." *Nature* 577 (7791): 519–25. <https://doi.org/10.1038/s41586-019-1916-6>.
- Imlay, James A. 2003. "Pathways of Oxidative Damage." *Annual Review of Microbiology* 57 (1): 395–418. <https://doi.org/10.1146/annurev.micro.57.030502.090938>.
- Imlay, James A. 2013. "The Molecular Mechanisms and Physiological Consequences of Oxidative Stress: Lessons from a Model Bacterium." *Nature Reviews Microbiology* 11 (7): 443–54. <https://doi.org/10.1038/nrmicro3032>.
- Lane, Nick, and William Martin. 2010. "The Energetics of Genome Complexity." *Nature* 467 (7318): 929–34. <https://doi.org/10.1038/nature09486>.
- Latour, Bruno. 1999. *Pandora's Hope: Essays on the Reality of Science Studies*. Harvard University Press.
- Lukács, Georg. (1923) 1971. *History and Class Consciousness: Studies in Marxist Dialectics*. Translated by Rodney Livingstone. MIT Press.
- Lynch, Michael, and Georgi K. Marinov. 2015. "The Bioenergetic Costs of a Gene." *Proceedings of the National Academy of Sciences* 112 (51): 15690–95. <https://doi.org/10.1073/pnas.1514974112>.
- Lyons, Timothy W., Christopher T. Reinhard, and Noah J. Planavsky. 2014. "The Rise of Oxygen in Earth's Early Ocean and Atmosphere." *Nature* 506 (7488): 307–15. <https://doi.org/10.1038/nature13068>.

- Mack, Arien, and Irvin Rock. 1998. *Inattentional Blindness*. The MIT Press. <https://doi.org/10.7551/mitpress/3707.001.0001>.
- Margulis, Lynn. 1967. "On the Origin of Mitosing Cells." *Journal of Theoretical Biology* 14 (3): 225–74. [https://doi.org/10.1016/0022-5193\(67\)90079-3](https://doi.org/10.1016/0022-5193(67)90079-3).
- Martin, William, and Miklós Müller. 1998. "The Hydrogen Hypothesis for the First Eukaryote." *Nature* 392 (6671): 37–41. <https://doi.org/10.1038/32096>.
- Marx, Karl. (1844) 1959. *Economic and Philosophic Manuscripts of 1844*. Translated by Martin Milligan. Progress Publishers.
- Mitchell, Peter. 1961. "Coupling of Phosphorylation to Electron and Hydrogen Transfer by a Chemi-Osmotic Type of Mechanism." *Nature* 191 (4784): 144–48. <https://doi.org/10.1038/191144a0>.
- Moody, Edmund R. R., Sandra Álvarez-Carretero, Tara A. Mahendrarajah, et al. 2024. "The Nature of the Last Universal Common Ancestor and Its Impact on the Early Earth System." *Nature Ecology & Evolution* 8 (9): 1654–66. <https://doi.org/10.1038/s41559-024-02461-1>.
- Muñoz-Gómez, Sergio A., Edward Susko, Kelsey Williamson, et al. 2022. "Site-and-Branch-Heterogeneous Analyses of an Expanded Dataset Favour Mitochondria as Sister to Known Alphaproteobacteria." *Nature Ecology & Evolution* 6 (3): 253–62. <https://doi.org/10.1038/s41559-021-01638-2>.
- Navarrete, Ana, Carel P. van Schaik, and Karin Isler. 2011. "Energetics and the Evolution of Human Brain Size." *Nature* 480 (7375): 91–93. <https://doi.org/10.1038/nature10629>.
- Ong, Walter J. 1982. *Orality and Literacy: The Technologizing of the Word*. Methuen.
- Ostrander, Chadlin M., Andy W. Heard, Yunchao Shu, et al. 2024. "Onset of Coupled Atmosphere–Ocean Oxygenation 2.3 Billion Years Ago." *Nature* 631 (8020): 335–39. <https://doi.org/10.1038/s41586-024-07551-5>.
- Parnas, D. L. 1972. "On the Criteria to Be Used in Decomposing Systems into Modules." *Communications of the ACM* 15 (12): 1053–58. <https://doi.org/10.1145/361598.361623>.
- Picard, Martin, and Bruce S. McEwen. 2018. "Psychological Stress and Mitochondria: A Systematic Review." *Psychosomatic Medicine* 80 (2): 141–53. <https://doi.org/10.1097/psy.0000000000000545>.
- Plato. 1995. *Phaedrus*. Translated by Alexander Nehamas and Paul Woodruff. Hackett.
- Polanyi, Michael. 1966. *The Tacit Dimension*. Doubleday.
- Raichle, Marcus E., and Debra A. Gusnard. 2002. "Appraising the Brain's Energy Budget." *Proceedings of the National Academy of Sciences* 99 (16): 10237–39. <https://doi.org/10.1073/pnas.172399499>.

- Rangaraju, Vidhya, Marcel Lauterbach, and Erin M. Schuman. 2019. "Spatially Stable Mitochondrial Compartments Fuel Local Translation During Plasticity." *Cell* 176 (1-2): 73–84.e15. <https://doi.org/10.1016/j.cell.2018.12.013>.
- Rensink, Ronald A., J. Kevin O'Regan, and James J. Clark. 1997. "To See or Not to See: The Need for Attention to Perceive Changes in Scenes." *Psychological Science* 8 (5): 368–73. <https://doi.org/10.1111/j.1467-9280.1997.tb00427.x>.
- Roebroeks, Wil, and Paola Villa. 2011. "On the Earliest Evidence for Habitual Use of Fire in Europe." *Proceedings of the National Academy of Sciences* 108 (13): 5209–14. <https://doi.org/10.1073/pnas.1018116108>.
- Russell, Andrew L., and Lee Vinsel. 2016. "Hail the Maintainers." *Aeon*, April 7. <https://aeon.co/essays/innovation-is-overvalued-maintenance-often-matters-more>.
- Schmandt-Besserat, Denise. 1992. *Before Writing, Vol. 1: From Counting to Cuneiform*. University of Texas Press.
- Sessions, Alex L., David M. Doughty, Paula V. Welander, Roger E. Summons, and Dianne K. Newman. 2009. "The Continuing Puzzle of the Great Oxidation Event." *Current Biology* 19 (14): R567–74. <https://doi.org/10.1016/j.cub.2009.05.054>.
- Simon, Herbert A. 1962. "The Architecture of Complexity." *Proceedings of the American Philosophical Society* 106 (6): 467–82. <https://www.jstor.org/stable/985254>.
- Simondon, Gilbert. (1958) 2017. *On the Mode of Existence of Technical Objects*. Translated by Cecile Malaspina and John Rogove. Univocal.
- Simons, Daniel J., and Christopher F. Chabris. 1999. "Gorillas in Our Midst: Sustained Inattentional Blindness for Dynamic Events." *Perception* 28 (9): 1059–74. <https://doi.org/10.1068/p2952>.
- Spang, Anja, Jimmy H. Saw, Steffen L. Jørgensen, et al. 2015. "Complex Archaea That Bridge the Gap Between Prokaryotes and Eukaryotes." *Nature* 521 (7551): 173–79. <https://doi.org/10.1038/nature14447>.
- Spolsky, Joel. 2002. "The Law of Leaky Abstractions." *Joel on Software*, November 11. <https://www.joelonsoftware.com/2002/11/11/the-law-of-leaky-abstractions/>.
- Star, Susan Leigh. 1999. "The Ethnography of Infrastructure." *American Behavioral Scientist* 43 (3): 377–91. <https://doi.org/10.1177/00027649921955326>.
- Star, Susan Leigh, and Karen Ruhleder. 1996. "Steps Toward an Ecology of Infrastructure: Design and Access for Large Information Spaces." *Information Systems Research* 7 (1): 111–34. <https://doi.org/10.1287/isre.7.1.111>.
- Stiegler, Bernard. (1994) 1998. *Technics and Time, 1: The Fault of Epimetheus*. Translated by Richard Beardsworth and George Collins. Stanford University Press.

- Street, Brian V. 1984. *Literacy in Theory and Practice*. Cambridge University Press.
- Tomasello, Michael. 1999. *The Cultural Origins of Human Cognition*. Harvard University Press.
- Vinsel, Lee, and Andrew L. Russell. 2020. *The Innovation Delusion: How Our Obsession with the New Has Disrupted the Work That Matters Most*. Currency.
- Weber, Max. (1919) 1946. "Science as a Vocation." In *From Max Weber: Essays in Sociology*, edited by H. H. Gerth and C. Wright Mills. Oxford University Press.
- Weiss, Madeline C., Filipa L. Sousa, Natalia Mrnjavac, et al. 2016. "The Physiology and Habitat of the Last Universal Common Ancestor." *Nature Microbiology* 1 (9): 16116. <https://doi.org/10.1038/nmicrobiol.2016.116>.
- Whitehead, Alfred North. 1925. *Science and the Modern World*. Macmillan.
- Wrangham, Richard. 2009. *Catching Fire: How Cooking Made Us Human*. Basic Books.
- Wrangham, Richard W., James Holland Jones, Greg Laden, David Pilbeam, and NancyLou Conklin-Brittain. 1999. "The Raw and the Stolen: Cooking and the Ecology of Human Origins." *Current Anthropology* 40 (5): 567–94. <https://doi.org/10.1086/300083>.